25ECPCC302IAE-II

Seat No. : 15

Student Name : Harsh

St. John College of Engineering and Management, Palghar,
Maharashtra
Autonomous Institute

Internal Assessment Examinations, October 2026

B.Tech. Electronics & Computer Science - SEMESTER III

25ECPCC302 - SDC

Time : 1 Hour **Maximum Marks : 10/20**

College of Engineering & Management SJCEM PALGHAR St. John

Instructions

- All Questions are compulsory and carry equal weightage of 1 marks
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Max. marks(5)	Course Outcome	BT Level
	a)Define Virtual ground Concept	1	CO4	L1
	b)Explain the different types of close loop configuration	1	CO4	L2
	c)Draw the input and output waveforms of Differentiator	1	CO5	L1
	d)What is the condition for summing and averaging amplifier	1	CO5	L2
	e)Write the applications of Monostable multivibrator	1	CO6	L1
	f)Write any two specifications of IC55 5	1	CO6	L2
Q2	a)Elaborate following parameters of OP AMP and state its ideal and practical value for IC 741.CMRR ,Slew Rate ,Open loop gain , PSRR	5	CO4	L3
	OR			
	b)Apply the block diagram of MOSFET differential amplifier to explain its operation.	5	CO4	L3
Q3	a)Distinguish linear and nonlinear applications of opamp	5	CO5	L4
	OR			
	b)Elaborate the concept and working of Zero crossing Detector .	5	CO5	L3

Q4	a)Model the circuit diagram of an Astable multivibrator using IC 555 for a frequency of two KHZ and duty cycle of 70%	5	CO6	L3
	OR			
	b)Develop a monostable 555 timer circuit to produce an output pulse of 20 second wide Draw the circuit diagram	5	CO6	L3